SIYAMTHANDA MAFUTA

Computer Engineering Graduate | Software, Networks & Intelligent Systems

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PROFESSIONAL SUMMARY

Computer Engineering student with a completed Diploma in Computer Engineering and four months of professional development and collaborative project experience through the CAPACITI Demand Academy. Experience spans software development, networking, databases, embedded systems and artificial intelligence, supported by academic and personal projects. Interested in opportunities to contribute and grow across software, IT infrastructure, networking and intelligent systems.

TECHNICAL SKILLS

- **Software & Development:** Python, Java, JavaScript, Node.js, Express, React, HTML, CSS, Backend Development
- **Networks & Systems:** TCP/IP, Routing & Switching, Cisco Networking, Linux, Network Troubleshooting
- **AI & Data:** Artificial Intelligence, Machine Learning Fundamentals, Generative AI, Prompt Engineering, Signal Processing
- **Databases & Engineering:** MySQL, MongoDB, Database Design, ESP32, Arduino, Embedded Systems, MATLAB, Git/GitHub

PROFESSIONAL DEVELOPMENT

Demand Academy Participant | CAPACITI | Cape Town **2026 - Present**

- Four months of professional development and collaborative project experience through the CAPACITI Demand Academy.
- Completed IBM courses in back-end development with Node.js and Express, and front-end development with React, alongside learning in AI, Python and machine learning.
- Collaborating on bootcamp projects and building a professional portfolio to showcase technical growth and project experience.

SELECTED PROJECTS

CSI-Heart - Contactless Vital Sign Extraction via Wi-Fi Sensing

Academic / Research Group Project | Project Team Member / Developer

Contributing to a system that explores Wi-Fi Channel State Information (CSI) for contactless vital-sign extraction using ESP32 devices, signal processing and machine-learning approaches.

RoadReady Traffic Services

CAPACITI / Demand Academy Collaborative Project | Project Team Member / Developer

Collaborative software project focused on digital traffic service requests and related operational workflows.

RoadReady Insights

CAPACITI / Demand Academy Collaborative Project | Project Team Member / Developer

Interactive platform for sharing traffic service experiences and insights.

InfraTrack System

Academic Group Project | Project Team Member / Developer

A full-stack ticket and infrastructure maintenance management system modelled around the City of Johannesburg Public Works Department workflow (academic prototype).

Duplicate File Finder

Personal Software Project | Individual Developer

Java-based application developed to identify and manage duplicate files.

EDUCATION

Bachelor of Engineering Technology in Computer Engineering

Cape Peninsula University of Technology (CPUT) | 2025 - 2026 (Currently Completing)

Diploma in Computer Engineering

Cape Peninsula University of Technology (CPUT) | 2020 - 2024 (Completed)

SELECTED CERTIFICATIONS

- Artificial Intelligence Bootcamp (AI) - CAPACITI / Coursera
- Google AI Essentials Specialization - Google
- Machine Learning: Regression and Classification - DeepLearning.AI
- Generative AI with Large Language Models - DeepLearning.AI / AWS
- Python for Data Science, AI & Development - IBM
- Introduction to Artificial Intelligence (AI) - IBM
- Developing Back-End Apps with Node.js and Express - IBM
- Developing Front-End Apps with React - IBM
- Additional CAPACITI learning: AI For Everyone, Prompt Engineering, Responsible AI and AI productivity courses